

Credit - Drishti Ias

Illegal Sand Mining in National Chambal Sanctuary



Source: HT

Why in News?

The **Supreme Court** recently expressed concern over **illegal sand mining in the National Chambal Sanctuary** and directed the governments of **Uttar Pradesh, Madhya Pradesh and Rajasthan** to take strict action, as mining activities are damaging the **Chambal river ecosystem and threatening key species such as gharials** .

Summary:

- The Supreme Court recently raised concerns over illegal sand mining in the National Chambal Sanctuary, highlighting its severe impact on the Chambal river ecosystem and endangered species such as the Gharial and the Ganges river dolphin.
- Illegal sand mining in India is driven by rising construction demand, governance failures, weak enforcement, and policy gaps, leading to ecological degradation, riverbank erosion, groundwater depletion, and threats to biodiversity and infrastructure.
- The judiciary has strengthened regulation through landmark judgments and environmental principles like the Public Trust Doctrine and Precautionary Principle, while measures such as digital surveillance, stronger governance, scientific mining practices, and promotion of M-Sand are crucial to curb illegal mining.

What is Sand Mining?

- **About Sand Mining:** Sand mining refers to the extraction of **sand and gravel from riverbeds, coasts and dunes mainly for construction activities**.
 - While sand is essential for infrastructure development, excessive and illegal mining beyond natural replenishment levels causes serious environmental and geological damage.
- **Key Drivers of Illegal Sand Mining:**

- **Economic Drivers:** India's rapid urbanization and infrastructure expansion have created a massive demand-supply gap for sand used in construction activities.
 - **Illegal sand mining remains highly profitable as miners avoid costs related to environmental clearances,** royalties and regulatory compliance.
- **Administrative and Governance Failures:** Illegal sand mining is often supported by a nexus between sand mafias, local politicians and corrupt officials, resulting in weak enforcement and poor accountability.
 - In addition, authorities frequently lack adequate **manpower, surveillance technology and monitoring capacity** to regulate mining activities effectively.
- **Policy and Technical Gaps :** Outdated or inaccurate District Survey Reports often fail to determine sustainable extraction limits, leading to excessive mining.
 - Weak transport monitoring systems and the slow adoption of alternatives such as **Manufactured Sand** further make it difficult to curb illegal sand mining.
- **Major Impacts of Illegal Sand Mining**
 - **Ecological Degradation:** Causes riverbank erosion, destruction of aquatic habitats and loss of biodiversity.
 - **Hydrological Disruption:** Alters river flow, deepens riverbeds, lowers groundwater levels and increases flood risks.
 - **Threat to Infrastructure:** Weakens bridges, embankments and nearby public infrastructure.
- **Regulation of Sand Mining in India:**
 - Under the **Mines and Minerals (Development and Regulation) Act, 1957** , sand is classified as a minor mineral, making its regulation primarily a State subject.
 - The **Environment Impact Assessment Notification, 2006** mandates prior environmental clearance for sand mining projects.
 - The **Sustainable Sand Mining Management Guidelines, 2016** promote scientific and sustainable mining practices through District Survey Reports.
 - The **Enforcement and Monitoring Guidelines for Sand Mining, 2020** emphasize stricter monitoring through drones, satellite tracking and barcoding of mineral transport vehicles.

Key Judgements Related to Sand Mining

- **Deepak Kumar v. State of Haryana, 2012:** The Supreme Court ruled that all sand mining activities, **including those on areas below 5 hectares**, must obtain environmental clearance to curb ecological degradation and illegal mining.
- **State of Uttar Pradesh v. Gaurav Kumar, 2025:** The Supreme Court ruled that a valid and finalized **District Survey Report (DSR) is absolutely mandatory for granting environmental clearance for sand mining operations.**
 - The Court established a "**No District Survey Report, No Sand-Mining**" rule, declaring that draft DSRs are legally invalid for auctioning or permitting mining

The judiciary has consistently relied on two key environmental principles while dealing with sand mining issues:

- **Public Trust Doctrine** : Evolved from **M.C. Mehta v. Kamal Nath , 1996** this principle states that natural resources such as rivers and sand belong to the public, and the State acts only as a trustee responsible for protecting them from excessive private exploitation.
- **Precautionary Principle** : Recognized in **Vellore Citizens Welfare Forum v. Union of India , 1996** it holds that the absence of complete scientific certainty cannot be used as a justification for delaying measures to prevent serious or irreversible environmental damage.

What Measures can be Adopted to Curb Illegal Sand Mining in India?

- **Technological Integration (Digital Surveillance)**
 - **Satellite & Drone Monitoring:** Periodic aerial surveys (using drones and satellite imagery) are essential to map riverbeds and detect unauthorized excavations in real-time. This reduces the need for manual patrolling and provides irrefutable evidence for legal action.
 - **e-Transit Passes & Bar-coding:** Implementing a mandatory, tamper-proof, QR-code-based transit pass system for every vehicle. This should be integrated with GPS tracking to monitor the vehicle's route from the source (mine site) to the destination.
 - **IT-based Monitoring Dashboards:** Districts should maintain a real-time portal (like the "Mining Surveillance System") that cross-references mineral output with the royalty paid, helping flag inconsistencies instantly.
- **Strengthening Administrative Governance**
 - **District Survey Reports (DSR) Reform:** DSRs must be scientifically robust, based on GIS mapping and annual replenishment studies. These reports should be made public to allow for social audits by civil society.
 - **Separation of Duties:** To break the "mafia-official" nexus, enforcement teams should be independent of the local mining department. Creating dedicated "Anti-Illegal Mining Task Forces" that report directly to the state level or a nodal officer can reduce local administrative pressure.
 - **Accountability of Financiers:** The focus must shift from penalizing low-level drivers to tracing the "money trail." Investigations should target the owners, financiers, and contractors behind the operations using the Prevention of Money Laundering Act (PMLA) where applicable.

- **Policy and Economic Incentives**

- **Promotion of M-Sand:** Actively incentivizing the production and use of Manufactured Sand (M-Sand). Governments can mandate the use of M-Sand in public sector construction projects and provide subsidies for establishing crushers, thereby reducing the demand for river-based sand.
- **Rationalizing Royalty and Lease Terms:** By streamlining the process for obtaining legal leases and reducing bureaucratic hurdles, the state can make legal mining more attractive and profitable than the "high-risk" illegal trade.
- **Community Participation:** Establishing "Village/River Committees" that are empowered to monitor riverbanks. Giving local communities a stake in sustainable resource management (similar to Joint Forest Management) can turn them into guardians of the river.

National Chambal Sanctuary

- The National Chambal Sanctuary is a tri-state protected area spread across Rajasthan, Madhya Pradesh, and Uttar Pradesh along the Chambal River.
 - Declared a protected area in 1978–79, it covers nearly 5,400 sq. km and is jointly managed by the three states.
- The sanctuary is renowned for its pristine river ecosystem, ravines, sandy banks, and rich biodiversity.
- It supports rare aquatic species such as **Gharials**, **Gangetic river dolphins** , **red-crowned roofed turtles** , **Indian skimmers** and **mugger crocodiles** .

Gharial



- Gharial is scientifically known as ***Gavialis gangeticus*** and is a **fish-eating crocodilian found mainly in freshwater river systems**.
 - **IUCN Status:** It is listed as **Critically Endangered** on the [IUCN Red List](#) .
- It is protected under **Schedule I** of the **Wildlife (Protection) Act, 1972** , which provides the highest level of legal protection.
- **Habitat:** It prefers deep, fast-flowing rivers with undisturbed sandy banks for basking and nesting, the **Chambal River** is one of its most important natural habitats in India.

Gangetic River Dolphin



- Gangetic River Dolphin is scientifically known as **Platanista gangetica** (locally known as " **Susu** ") and is a freshwater dolphin found mainly in the Ganga-Brahmaputra-Meghna river system.
 - **IUCN Status:** It is listed as **Endangered** on the IUCN Red List.
 - It is also protected under **Schedule I** of the Wildlife (Protection) Act, 1972.
- **Habitat:** It prefers deep river channels, confluences and meandering river stretches with adequate water flow. It is also India's **National Aquatic Animal** .

Conclusion:

Illegal sand mining in the National Chambal Sanctuary threatens the fragile river ecosystem and endangered species like the Gharial and the Ganges river dolphin. Strong enforcement and sustainable conservation measures are essential to protect this unique natural heritage.

Drishti Mains Question

Discuss the ecological significance of riverine species such as the Gangetic River Dolphin and Gharial. How do illegal sand mining and river pollution threaten their survival?

Frequently Asked Questions (FAQs)

1. Why is the Chambal river important for wildlife?

The Chambal river supports important aquatic species such as gharials, Gangetic river dolphins, Indian skimmers, red-crowned roofed turtles and mugger crocodiles.

2. How does sand mining affect gharials?

Sand mining destroys natural sandbanks used by gharials for nesting. It also disturbs breeding sites and can destroy eggs laid in the sand.

3. Why are Indian skimmers vulnerable to disturbance?

Indian skimmers lay their eggs directly on shallow sandbanks. If disturbed by human activity, they may leave the eggs unattended, causing them to dry up.

4. What enforcement gaps were noted by the Supreme Court?

The Court pointed to unregistered tractors carrying illegally mined sand. It also noted inadequate protection for forest staff, shortage of guards and the need for stronger enforcement against organised sand mining networks.

5. Which States are associated with the National Chambal Sanctuary?

The National Chambal Sanctuary spans Madhya Pradesh, Rajasthan and Uttar Pradesh.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/Y-P6wXPJTgc](https://www.youtube.com/embed/Y-P6wXPJTgc)]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. If you want to see gharials in their natural habitat, which one of the following is the best place to visit? (2017)

- (a) Bhitarkanika Mangroves
- (b) Chambal River
- (c) Pulicat Lake
- (d) Deepor Beel

Ans: (b)

Mains

Q. What are the consequences of illegal mining? Discuss the Ministry of Environment and Forests' concept of "GO AND NO GO" zones for coal mining. (2013)

150 Years of Indian Association for the Cultivation of Science



Source: PIB

The **Indian Association for the Cultivation of Science (IACS)** commemorated 150 years of its establishment.

- The government also inaugurated the indigenously built **Plasma Enhanced Chemical Vapor Deposition (PECVD)** system at IACS for fabrication of **India's first amorphous silicon solar cell and RETINA** (Research Entrepreneurship for Translation, Innovation and Navigation), a centre aimed at linking fundamental research with entrepreneurship, startups and societal applications.
- **About Indian Association for the Cultivation of Science(IACS):**
 - IACS was founded in **1876** by **Dr. Mahendralal Sircar** . It is regarded as the **first research institution in Asia established by Indians** for the cultivation of modern science through original research.
 - **Pandit Iswar Chandra Vidyasagar** and **Kesab Chandra Sen** were among the members of the first Trustee Board of IACS, with Dr. Sircar himself as the first Honorary Secretary
 - IACS became associated with several leading Indian scientists, including **Jagadish Chandra Bose, Meghnad Saha, S.N. Bose and Sir C.V. Raman** .
 - Sir **C.V. Raman** carried out his work on the **Raman Effect** at IACS. This discovery earned him the **Nobel Prize in Physics in 1930** , making it **India's first Nobel Prize in Science**.
 - The **Raman Effect** is the inelastic scattering of light by molecules, resulting in a shift in the wavelength and energy of the scattered photons.
 - When monochromatic light strikes a substance, most photons scatter elastically with no change in energy, but a tiny fraction exchange energy with the molecules, emerging with different wavelengths

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